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Predictive Model for Asthma Diagnosis Using Advanced Feature Selection and Voting Techniques with Model Explainability

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***Abstract*— Millions of people worldwide suffer from asthma, a chronic respiratory disease that frequently needs precise diagnosis and treatment to enhance patient outcomes. The proposed work uses a hard voting ensemble model to predict asthma using a machine learning based method that achieves a highest accuracy of 93.83%. SMOTE is used in the suggested pipeline to alleviate class imbalance and extra trees classifier (ETC) with recursive feature elimination (RFE) is incorporated for efficient feature selection. The prediction performance was improved by training and aggregating a variety of machine learning models employing soft and hard voting such as random forest, decision tree, k-nearest neighbors, gradient boosting and XGBoost. The model's interpretability was assessed using SHAP analysis which revealed important variables influencing asthma predictions. This methodology offers a promising tool for supporting clinical decision making and optimizing the allocation of healthcare resources because of its exceptional efficacy, scalability and explainability.**

Keywords— XGBoost, ETC, RFE, Secyu, Dryum

I. INTRODUCTION

Asthma is a chronic inflammatory condition of the airways that affects millions of individuals worldwide, regardless of age, gender or socioeconomic background [1]. It is typified by inflammation, blockage and hyperresponsiveness of the airways which show up as coughing, chest tightness, wheezing and dyspnea [2]. Asthma is a very complex illness that requires tailored management and treatment strategies because the frequency and severity of these symptoms can vary substantially. There are environmental and genetic factors that contribute to asthma. While lifestyle variables including smoking, exposure to air pollution, occupational risks and respiratory infections play important roles in its onset and exacerbation, family history, genetic susceptibility and early life exposure to allergens are important risk factors [3]. The dynamics of the disease are further complicated by environmental triggers such as mold, dust mites, pollen, animal dander and cold air as well as by stress and physical exercise. Significant difficulties in diagnosis, treatment and prevention arise from this intricacy.

Asthma affects more than 300 million people worldwide and its incidence is rising significantly, especially in urban and industrial areas [4].Urban living circumstances, environmental changes and greater exposure to both indoor and outdoor contaminants are all factors contributing to this increase. Asthma causes a lot of morbidity, especially in children but it also costs a lot of money because of medical expenses, lost productivity and a worse standard of living for both patients and caregivers. Asthma is a serious public health issue since severe flare-ups can result in hospitalization and in extreme circumstances death. Medical research has greatly advanced our understanding of asthma which has resulted in the creation of novel treatment alternatives such as biologics, inhaled corticosteroids and tailored therapeutic techniques [5]. Furthermore, because of the intricate interactions between heredity, environment and immune system reactions, asthma requires constant research to identify the underlying mechanisms, enhance diagnostic methods and provide more potent treatments.

The point below is the contribution of proposed work.

- ETC based RFE for feature selection and SMOTE for class imbalance were to provide balanced relevant data.

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- Several models (such as random forest, catboost and XGBoost) were trained and ensemble learning improved performance.
- SHAP was used to examine feature contributions and enhance transparency.
- Created a methodical, comprehensive approach for practical machine learning applications.

II. Related Work

Vatsal et al. [6] using data from 29,396 patients, ensemble machine learning models that use algorithms like random forests and gradient boosting have reached 90% accuracy. These models have promise for better outcomes and individualized asthma care, despite implementation and data quality issues. Joverlyn Gaudillo et al. [7] random forest (RF) was proven to be an excellent feature selection method. RF-SVM achieved an accuracy of 62.5% and an AUC of 0.62. These findings demonstrate the potential of machine learning to predict genetic susceptibility to complicated diseases like asthma.

Julie L. Harvey et al. [8] employing machine learning classifiers discovered that the RF model obtained 90.9% accuracy, with major variables including sex, difficulty breathing, allergies and medications. These findings demonstrate ML's potential for early asthma prediction in clinical settings. Dejing Li et al. [9] used affinity graph enhanced classifier (AGEC) enhances prediction by capturing sample relationships through blood marker characteristics. AGEC obtained 72.5% accuracy, surpassing models such as SVM (69.8%) and FWAdaBoost (61.0%), indicating its promise for improved asthma prediction. Karin Lisspers et al. [10] employed machine learning examined data from 29,396 patients to predict exacerbations within 15 days. Comorbidity burden and history of exacerbations were important predictors. However, model validation demonstrated poor performance (AUPRC = 0.007), implying that past clinical data alone may not be sufficient for effective short term risk prediction.

Yao Tong et al. [11] used data from 31,724 adult outpatients, this study created a machine learning model to predict the continuity of care (COC) for asthma patients. The model achieved an accuracy of 88.20%, AUC of 0.96, and F1 score of 0.86. Age, insurance status, comorbidities and the severity of asthma were important determinants of COC. The results provide a basis for future optimization and use in healthcare management by highlighting the potential of predictive modeling to improve clinical decision making, resource allocation and patient outcomes. Rebecca Howard et al. [12] used identification of asthma subtypes, especially in children has advanced because of machine learning approaches like latent class analysis. These revelations open the door to innovative medicines, tailored treatments and accurate preventative measures.

Seyed Vahid Razavi Termeh et al. [13] conducted in Tehran that mapped asthma prone areas using a RF model found that the highest incidence occurred in the fall with AUC scores ranging from 0.823 to 0.83 across seasons. This emphasizes how important spatial temporal modeling is for studying asthma. Mindy K. Ross et al. [14] used the predictor Pursuit (PP) method was examined and four phenotypes depending on pharmaceutical response were found. When it came to predicting asthma control, PP performed better than traditional models ($P < 0.001$) with serum eosinophils and bronchodilator responsiveness serving as important predictors. This demonstrates how ML may be used to customize asthma treatment. Matthew Xie et al. [15] were created using NHIS data and the best results were obtained with logistic regression (AUC 0.7654). Parental asthma and low family income were important risk factors, confirming ML's function in early asthma identification.

III. Methodology

In Fig. 1 this flowchart depicts an organized machine learning pipeline that begins with asthma dataset and proceeds through preprocessing and modeling phases to ensure optimal performance and interpretability. The procedure begins with the use of SMOTE (synthetic minority oversampling) to rectify class imbalance in the data. Recursive feature elimination (RFE) and an extra trees classifier (ETC) are then used for feature selection in order to determine which features are most crucial for training. The processed dataset is divided into two parts: 20% for testing and 80% for training in order to ensure a reliable assessment. The prepared training data is used to train a variety of machine learning models such as the random forest, decision tree, KNN, gradient boosting, CatBoost, XGBoost, and extra tree classifier. The predictions of these models are combined in an ensemble approach to improve generalization, accuracy and robustness. The performance of the ensemble model is assessed on the test set using important metrics like accuracy, recall, precision, f1-score, and AUC score. Lastly, the model's predictions are interpreted using SHAP (shapley additive explanations) which provides comprehensive insights into the role and impact of various characteristics. This improves the model's usability, transparency and trust in practical applications.

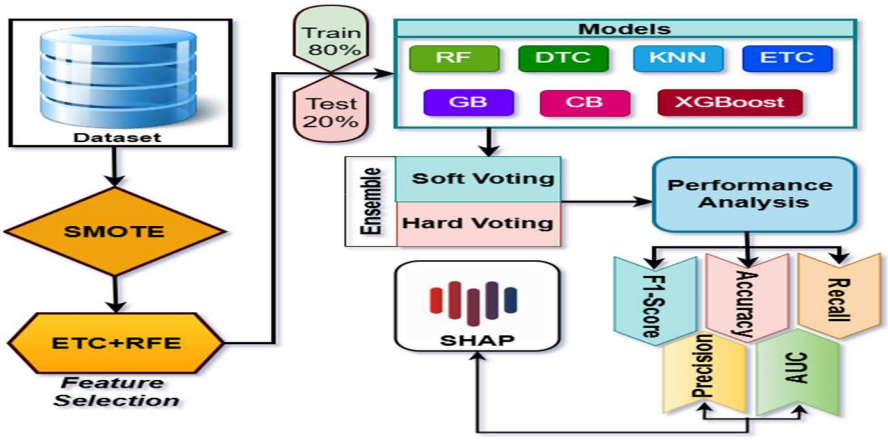


Fig. 1. Workflow of the proposed model.

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A. Dataset

The dataset was obtained from Kaggle [16]. It contains 2,392 patients, evenly divided by gender and ranging in age from 5 to 100+. Sleep problems (60.2%), wheezing (59.6%) and shortness of breath (50.0%) are common symptoms with a family history of asthma reported by 30.1%. Only 5.2% of people are diagnosed with asthma despite these criteria which provide valuable information on symptoms and risk factors.

B. Feature Selection

The process of feature selection employs recursive feature elimination (RFE) to repeatedly eliminate less significant features and extra trees classifier (ETC) to rank features according to their importance [17]. RFE refines the feature set by gradually removing the least significant variables whereas ETC uses ensemble decision trees to identify essential factors. By ensuring that only the most predictive and pertinent features are kept, this combination lowers dimensionality and enhances machine learning model performance. Select the feature according to the TABLE I.

TABLE I. FEATURE SELECTION BY ETC AND RFE

Feature	ET Importance	Selected by RFE
Age	0.170863	Yes
Gender	0.138644	No
Smoking	0.115782	No
SleepQuality	0.096832	Yes
DustExposure	0.066923	Yes
PetAllergy	0.066075	No
FamilyHistoryAsthma	0.064612	No
HayFever	0.052519	Yes
Wheezing	0.051191	No
ShortnessOfBreath	0.048215	No
ChestTightness	0.04747	Yes
Coughing	0.042639	Yes
NighttimeSymptoms	0.038234	Yes

In Fig. 2 the top ten features in the predictive model are highlighted with special attention paid to the seven features chosen by RFE which are shown by the blue bars. These comprise modest contributors like hay fever, chest tightness, coughing and nighttime symptoms as well as highly influential features like age, sleep quality and dust exposure all of which improve the model's overall performance. Some significant characteristics, such as smoking and gender (the second most important) were not chosen by RFE, most likely because they were redundant with other predictors. A family history of asthma and wheeze, among other unselected characteristics, also shown a moderate level of significance.

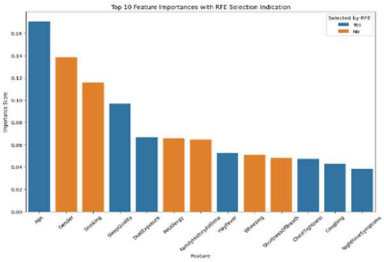


Fig. 2. RFE Identifies Key Features of Asthma Prediction Model.

C. Data Split

A great approach is to divide the data into two sets: 80% for training and 20% for testing, ensuring enough data for training while keeping a distinct set for evaluation. This divide enables the model to train well while providing an unbiased measure of its performance on previously unknown data, resulting in robust and dependable outcomes.

D. Extra Tree Classifier

It generates a number of trees, each with a unique random subset of the training set's attributes and data. Extra trees classifier increases unpredictability over traditional decision trees or random forests by randomly selecting split points inside each feature [18]. This increased unpredictability adds to improved generalization and less variation.

E. CatBoost

This gradient boosting technique requires little preparation such as label or one-hot encoding and performs well with categorical data. The main innovation of CatBoost is its native handling of categorical variables which enhances model performance and reduces the likelihood of overfitting. It employs a method known as ordered boosting, which is designed to prevent prediction leakage and speed up training using GPU acceleration [19]. Additionally, because of its stability and adaptability, CatBoost is a reliable choice for both regression and classification tasks. It often provides high accuracy with minimal effort in feature engineering. The learning problem's goal is to develop a function that minimizes the expected loss given by equation (1).

$$F(H) = \varphi L(y, H(X)) \tag{1}$$

IV. RESULT AND DISCUSSION

Accuracy measures overall correctness, precision demonstrates positive prediction accuracy and recall assesses the capacity to discern real positives. The F1 Score balances precision and recall whereas the ROC-AUC measures classification performance over many thresholds [20]. The framework uses high-performance CPUs to handle large datasets and train complex machine learning models. We determined the performance parameter by using the following equations: (2), (3), (4), (5), and (6).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Sensitivity = \frac{TN}{TN + FP} \tag{4}$$

$$F_1 - Score = 2 * \left(\frac{Sensitivity * Precision}{Sensitivity + Precision}\right) \tag{5}$$

$$Auc = \sum_{i=1}^{n-1} (FPR_{i+1} - FPR_i).\left(\frac{TPR_{i+1} + TPR_i}{2}\right) \tag{6}$$

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A. Confusion Metrics

In Fig. 3 the performance of different models and ensemble voting techniques for asthma prediction is compared in the confusion matrices. The most resilient method is hard voting (ETC + CB) (e) which produces the greatest results with 425 true negatives (TN), 427 true positives (TP) and few misclassifications (37 false positives [FP] and 19 false negatives [FN]). Soft voting (ETC + XGBoost) (c) provides a compromise between accuracy and error rates (414 TN, 427 TP, 48 FP, 19 FN) whereas XGBoost (a) also performs well with 417 TN, 425 TP and low errors (45 FP, 21 FN). Compared to the ensemble models, gradient boosting (b) exhibits greater misclassification rates (98 FP, 43 FN) whereas hard voting (GB + CB) (f) and soft voting (GB + CB) (d) yield respectable results but have marginally higher misclassification rates than the top performing models. Overall, ensemble approaches hard voting in particular perform better in terms of prediction by utilizing the advantages of individual models.

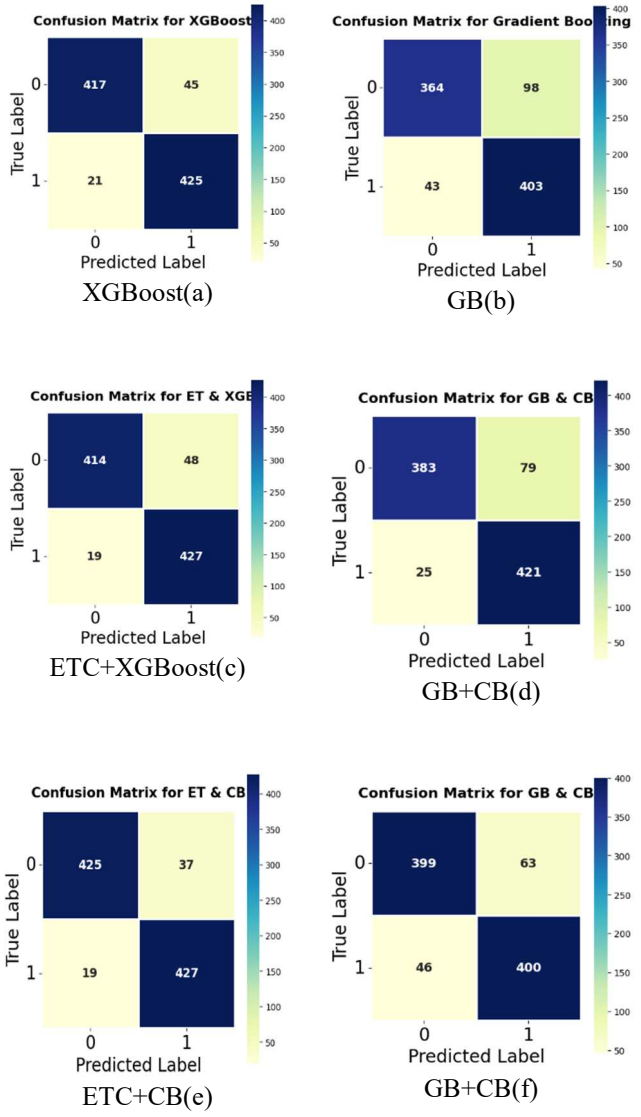


Fig. 3. Confusion metrics for ML and Ensemble models.

B. Figures and Tables

In TABLE II the performance scores of machine learning models with feature selection are displayed. Predictive performance is best achieved with XGBoost (92.73% accuracy, 97.58% AUC) and extra tree (92.51% accuracy, 97.94% AUC). Then, random forest (90.64% accuracy, 97.35% AUC) and CatBoost (90.97% accuracy, 97.01% AUC). The poor performance gradient boosting (84.47% accuracy, 90.66% AUC) performs the worst, followed by KNN (86.67% accuracy, 95.42% AUC) and decision tree (86.89% accuracy, 86.93% AUC).

TABLE II. MACHINE LEARNING MODEL PARAMETERS SCORE WITH FEATURE SELECTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	ROC (%)	AUC (%)
Random Forest	90.64	90.89	90.64	90.63	90.7	97.35
Decision Tree	86.89	87	86.89	86.89	86.93	86.93
K-Nearest Neighbors	86.67	88.62	86.67	86.53	86.87	95.42
Extra Trees	92.51	92.79	92.51	92.5	92.58	97.94
Gradient Boosting	84.47	85.02	84.47	84.43	84.57	90.66
CatBoost	90.97	91.72	90.97	90.94	91.08	97.01
XGBoost	92.73	92.86	92.73	92.73	92.78	97.58

In TABLE III (ET + XGB) achieved the greatest accuracy (92.62%) and AUC (98.27%), followed by CB + XGB (92.51%, 97.71%) and (RF + XGB) (92.40%, 97.94%), according to the table that highlights soft voting performance. While (GB + CB) achieves the lowest accuracy (88.55%) and AUC (95.09%), (ET + CB) also performs well (91.85% accuracy, 98.18% AUC). The power of these hybrid techniques in asthma prediction is demonstrated by the fact that combinations comprising (ET + XGB) consistently produce the best predictive performance across measures.

TABLE III. APPLY SOFT VOTING ON VARIOUS MODEL WITH FEATURE SELECTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	ROC Score (%)	AUC Score (%)
GB+CB	88.55	89.12	88.55	88.52	88.65	95.09
ET+XGB	92.62	92.8	92.62	92.62	92.68	98.27
RF+XGB	92.4	92.58	92.4	92.4	92.45	97.94
CB+XGB	92.51	92.93	92.51	92.5	92.59	97.71
GB+XGB	91.3	91.69	91.3	91.29	91.38	96.51
ET+CB	91.85	92.34	91.85	91.84	91.94	98.18
ET+GB	90.31	90.82	90.31	90.29	90.4	96.70

In TABLE IV the performance of hard voting result model combinations is summarized in the table. The best performance was obtained by (ET + CB) (93.83% accuracy, 93.91% precision, and 93.87% ROC score) which was closely followed by (ET + XGB) (93.72% across metrics). Strong results are also obtained with (RF + XGB) (92.51% accuracy) and (CB + XGB) (92.73% accuracy). (GB + XGB) (89.76%) and (ET + GB) (89.54%) exhibit moderate performance whilst (GB + CB) exhibits the lowest performance (88% accuracy, 88.02% ROC score). Overall, combinations of extra trees (ET) and CatBoost (CB) show greatest predictive power in hard voting ensembles. It is the highest accuracy on our proposed model.

TABLE IV. APPLY HARD VOTING ON VARIOUS MODEL WITH FEATURE SELECTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	ROC (%)
GB + CB	88	88.06	88	88	88.02
ET+XGB	93.72	93.72	93.72	93.72	93.72
RF+XGB	92.51	92.51	92.51	92.51	92.5

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CB+XGB	92.73	92.78	92.73	92.73	92.76
GB+XGB	89.76	89.8	89.76	89.75	89.73
ET+CB	93.83	93.91	93.83	93.83	93.87
ET+GB	89.54	89.54	89.54	89.54	89.52

The results of hard and soft voting in TABLE III and TABLE IV differ because they use different aggregating techniques. When individual models are robust and reliable, hard voting which chooses the class predicted by the majority of models can be more successful since it treats each vote equally. On the other hand, confident models are given more weight when soft voting averages expected probabilities which may not always coincide with the majority result. Given that most models in our study correctly predicted the class, hard voting performed better, resulting in a more reliable and accurate final judgment.

In Fig. 4 the chart shows the accuracy of individual models using the Extra Trees-RFE hybrid with seven features. The maximum accuracy is achieved by XGBoost (92.73%) which is followed by random forest (90.64%), extra tree (92.51%) and CatBoost (90.97%). K-nearest neighbors (86.67%) and decision trees (86.89%) have moderate accuracy whilst gradient boosting has the lowest performance (84.47%).

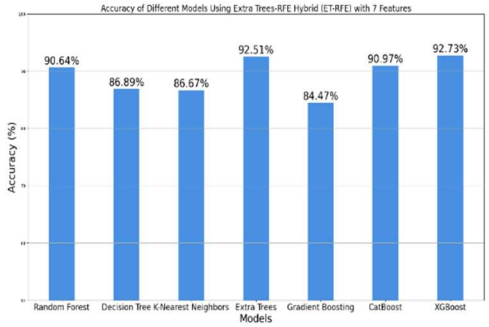


Fig. 4. Seven models accuracy score.

In Fig. 5 displays the accuracy ratings of the top seven combinations of soft voting models. The highest accuracy is 92.62% for ET & XGB which is closely followed by 92.51% for CB & XGB and 92.40% for RF & XGB. GB & XGB and ET & CB exhibit a modest level of performance with respective accuracies of 91.30% and 91.85%. The combination with the lowest accuracy, GB & CB is 88.55% while ET & GB attains 90.31%.

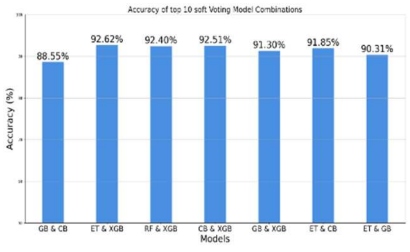


Fig. 5. Top seven accuracy score in soft voting.

In Fig. 6 displays the top seven hard voting model combinations accuracy scores. The maximum accuracy, 93.83%, is attained by ET & CB, closely followed by ET & XGB, 93.72%. RF & XGB and CB & XGB both exhibit strong performance, with respective accuracies of 92.51% and 92.73%. The accuracy of GB & CB is the lowest at 88.00%, while GB & XGB (89.76%) and ET & GB (89.54%) exhibit moderate results. The results show that hard voting combinations that use CatBoost (CB) and extra trees (ET) perform best.

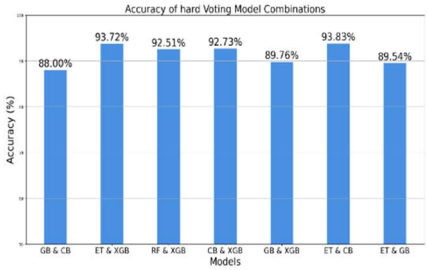


Fig.6. Top seven accuracy score in hard voting.

In TABLE V shows the suggested work performs in terms of asthma prediction accuracy when compared to earlier research. The suggested model outperforms all of the others with an impressive 93.83% accuracy while Vatsal et al. [6] and Julie L. Harvey et al. [8] achieved accuracies of 90.00% and 90.9%, respectively and other studies like Dejing Li et al. [9] and Joverlyn Gaudillo et al. [7] reported significantly lower accuracies (72.5% and 62.5%, respectively). This illustrates the strength and efficiency of our strategy which makes use of cutting-edge techniques like hard voting and feature selection using Extra Trees-RFE. The proposed w establishes a new standard in asthma prediction by reaching the maximum accuracy with a dataset of 2,392 patients, making it the most dependable and efficient solution when compared to previous studies.

TABLE V. PERFORMANCE ANALYSIS WITH OTHER WORK

Reference	Dataset	Accuracy (%)
Vatsal et al. [6]	Data Collected from Kaggle of 2,396 patients'	90.00
Joverlyn Gaudillo et al. [7]	Data collected from SNP database	62.5
Julie L. Harvey et al. [8]	Data collected from NSCH	90.9
Dejing Li et al. [9]	Data collected from ASHSTCM of 152 patients'	72.50
Proposed work	**Data Collected from Kaggle of 2,392 patients'**	93.83

C. Interpretability

In Fig. 7 the SHAP summary graphic shows different features contribute to a model's prediction. The impact of each feature is shown by horizontal bars. Coughing (-1.53) and nighttime symptoms (-1.42) decrease the probability of the expected outcome but positive SHAP values, including age (+1.71) and chest pressure (+1.49) show elements pushing the prediction in the direction of the good outcome. Hay fever (+0.48) had less of an impact than other features like dust exposure (-0.71) and sleep quality (+0.82). The combined contributions of these features, beginning with the base value of -0.243 are reflected in the final prediction output of f(x) = 0.61. Fig. 7 and 8 show the SHAP analysis, which shows major predictors of asthma, such as age, chest tightness, coughing, sleep quality, nighttime symptoms, dust exposure, and hay fever. These traits are consistent with recognized clinical knowledge and asthma pathogenesis. For example, chest tightness is a sign of bronchoconstriction, whereas coughing, particularly at night, is an early warning of airway

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inflammation. GINA (Global Initiative for Asthma) guidelines include sleep disruptions and nocturnal symptoms as diagnostic criteria, while environmental triggers such dust exposure and allergic rhinitis (hay fever) are known risk factors for asthma development or exacerbation. Age is also clinically significant since asthma symptoms and phenotypes differ between life stages. Beyond global insights, SHAP allows for patient-specific interpretation; for example, if a high-risk prediction is based on poor sleep and dust exposure, despite the absence of wheezing, physicians can still take preventive measures. This clinically grounded explainability improves the transparency and trustworthiness of our model, making it a useful tool for medical professionals in asthma diagnosis and risk stratification.

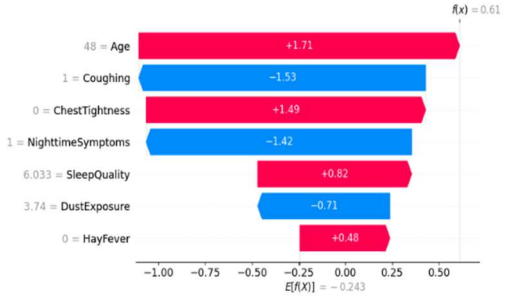


Fig. 7. SHAP Analysis Key Features Contributing to Model Predictions.

In Fig. 8 SHAP summary bar plot ranks the features according to their overall influence on the model's predictions by displaying the mean absolute SHAP values of each feature. Coughing (+1.17), chest tightness (+1.22) and age (+1.33) have the most effects. Hay fever (+0.71), dust exposure (+0.93), sleep quality (+0.83) and symptoms at night (+0.96) are additional significant characteristics.

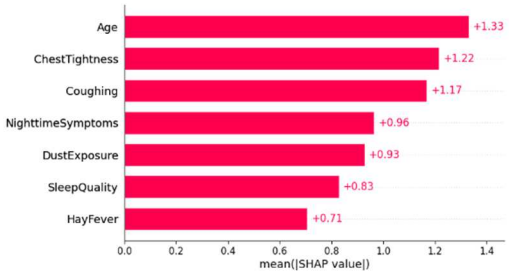


Fig. 8. SHAP Based Feature Importance Analysis for Asthma Prediction.

V. Conclusion

This study uses a hard voting ensemble method to achieve a high accuracy of 93.83%, demonstrating an efficient and interpretable machine learning pipeline for asthma prediction. The pipeline assures accuracy and generalizability by using SMOTE to address data imbalances, utilizing ETC with RFE for feature selection and integrating several reliable models. The model's transparency is further improved by using SHAP which qualifies it for real world use in healthcare settings. This method lays the groundwork for creating scalable and explicable predictive models for clinical decision making in addition to demonstrating the potential of ensemble approaches in disease prediction.

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